

* when you add water to anhydrous cobalt (II) chloride, the color changes from blue to pink

* when you add water to anhydrous copper (II) sulfate, the color changes from white to blue

Purity of water

- pure substances melt/boil at specific & sharp temperatures
- water (pure) has a boiling point of 100°C and melting point of 0°C
- impurities tend to increase the boiling point and decrease the melting point

distilled water:- water that has been heated to form a vapor and then condensed back to a liquid

- ↳ contains very few impurities
- ↳ is used in labs for practical chemistry
- ↳ tap water contains more impurities

* water from natural sources may contain dissolved oxygen, metal compounds, plastics, sewage, harmful microbes, nitrates from _____, phosphates from fertilisers & detergents.

beneficial	harmful
• dissolved oxygen for aquatic life	• some metal compounds are toxic
• metal compounds which provide essential minerals for life	• plastics harm aquatic life
	• sewage contains harmful microbes that spread disease
	• nitrates & phosphates lead to deoxygenation of water and damage to aquatic life

→ untreated water contains soluble/insoluble impurities
to purify the water:-

- ① water is pumped to sedimentation tanks
- ② mud, sand and other particles will fall to the bottom, and form a layer of sediment (sedimentation)
- ③ the smaller impure particles are then passed through sand & gravel filters that trap solid particles (filtration)
- ④ the water then can pass through carbon to remove tastes and odours.
- ⑤ Lastly a careful addition of chlorine is mixed with the water to get rid of bacteria and other microbes

* ammonium salts and nitrates are used as fertilisers.

NPK Fertilisers

• nitrogen, phosphorus & potassium

- ↳ nitrogen makes chlorophyll, protein & promotes healthy leaves
- ↳ potassium promotes growth & healthy fruit and flowers
- ↳ phosphorus promotes healthy roots

clean, dry air consists of:-

78% nitrogen (N_2), 21% oxygen (O_2), and the remainder is a mixture of noble gases and carbon dioxide (CO_2)

Source of air pollutants:-

CO_2 = complete combustion of carbon-containing fuels

CO = incomplete combustion of carbon-containing fuels

CH_4 = decomposition of vegetation & waste gases from digestion in animals

oxides (N) = car engines

SO_2 = combustion of fossil fuels that contain sulfur

the effect

CO_2 = increases global warming

CO = toxic gas

CH_4 = increases global warming

oxides (N) = acid rain, photochemical, smog & respiratory problems

SO_2 = acid rain

Greenhouse effect

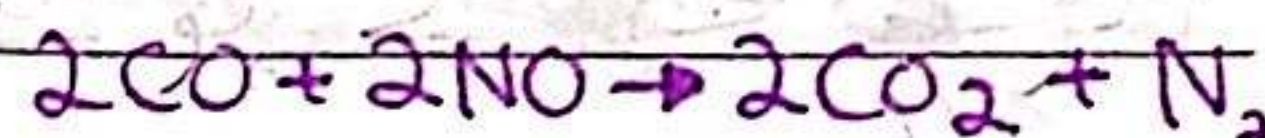
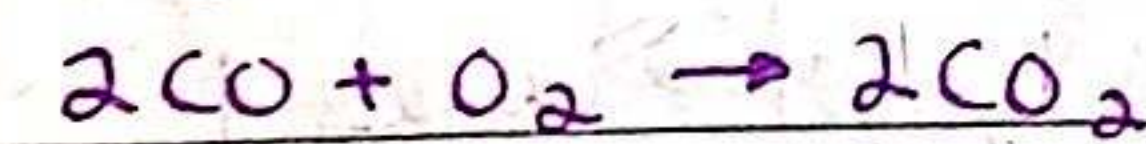
- the earth re-emits the radiation it receives from the sun but most of the radiation is absorbed by the greenhouse gases and reflected back at the surface because of which our planet is hotter than usual
- less energy is given out into space

Reducing environmental issues

climate change: plant trees, reduce livestock farming, decrease use of fossil fuels, increase use of hydrogen & renewable energy e.g. wind

acid rain: use of catalytic converters in vehicles, reducing emissions of SO_2 by using low-sulfur fuels and flue gas desulfurization with calcium oxide

- oxides of nitrogen are made when nitrogen & oxygen react in high pressure/temperature of the internal combustion engine and blast furnaces.



in the presence of
chlorophyll & light

Photosynthesis

carbon dioxide + water $\rightarrow$ glucose + oxygen

